

Deep Generative Models

Chapter 6: Generation by Diffusion Process

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Later Challenges

It turns out that *this approach does not lead to a **stable solution***

- ① *The score estimate is not accurate*
 - ↳ The model is trained by **extremely** noisy samples
 - ↳ The model has **limited capacity** as compared to **Bayes optimal**
- ② *The reverse trajectory is a first-order approximation*

$$x(t + dt) \approx x(t) + dx(t)$$

but to be accurate we should write

$$x(t + dt) = x(t) + dx(t) + \frac{1}{2}d^2x(t) + \dots$$

- ↳ We **cannot access** $d^j x(t)$ for $j > 1$ as **it is an SDE**
- ↳ Higher differentials can get **very large** due to **Brownian motion**

Alternative Look: *Diffusion as Markov Chain*

As mentioned earlier: we could see forward diffusion as a *Markov chain*

$$x_0 \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_{t-1} \longrightarrow x_t \longrightarrow \cdots \longrightarrow x_T$$

We know mathematically that the *reverse chain can exist*

$$x_0 \longleftarrow x_1 \longleftarrow \cdots \longleftarrow x_{t-1} \longleftarrow x_t \longleftarrow \cdots \longleftarrow x_T$$

Maybe we could *directly* learn it: in the forward process we have

$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\varepsilon_t$$

So, we could say the forward Markov chain is

$$Q(x_t|x_{t-1}) \equiv \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, 1 - \alpha_t)$$

Alternative Look: *Reverse Diffusion*

$$x_0 \longleftarrow x_1 \longleftarrow \cdots \longleftarrow x_{t-1} \longleftarrow x_t \longleftarrow \cdots \longleftarrow x_T$$

Now the question is: *what is the **reverse** Markov chain*

$$P(x_{t-1}|x_t)$$

which takes from distribution P_t to distribution P_{t-1} ?

Computational Solution

We consider a computational model $P_{\mathbf{w}}$

$$P_{\mathbf{w}}(x_{t-1}|x_t, t) = F_{\mathbf{w}}(x_{t-1}, t)$$

and try to find a way to train for mimicking the reverse trajectory

A Deep Look at *Forward Diffusion*

$$x_0 \xrightarrow{\alpha_1} x_1 \xrightarrow{\alpha_2} \cdots \longrightarrow x_{t-1} \xrightarrow{\alpha_t} x_t \xrightarrow{\alpha_{t+1}} \cdots \xrightarrow{\alpha_T} x_T$$

What does happen in forward process?

$$x_1 = \sqrt{\alpha_1}x_0 + \sqrt{1 - \alpha_1}\varepsilon_1$$

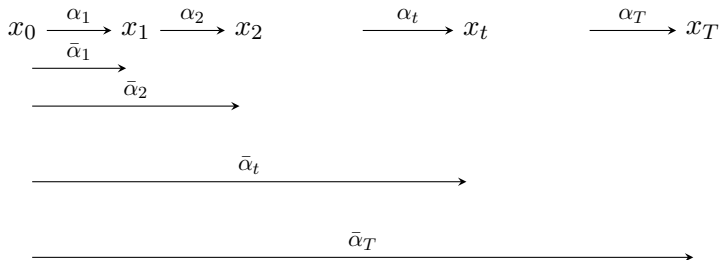
$$x_2 = \sqrt{\alpha_2}x_1 + \sqrt{1 - \alpha_2}\varepsilon_2 = \sqrt{\alpha_1\alpha_2}x_0 + \underbrace{\sqrt{\alpha_2}\sqrt{1 - \alpha_1}\varepsilon_1 + \sqrt{1 - \alpha_2}\varepsilon_2}_{\sqrt{1 - \alpha_1\alpha_2}\bar{\varepsilon}_2}$$

$\vdots$

$$x_t = \sqrt{\prod_{i=1}^t \alpha_i}x_0 + \sqrt{1 - \prod_{i=1}^t \alpha_i}\bar{\varepsilon}_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\bar{\varepsilon}_t$$

Direct Forward Links

We could also describe it with direct links from x_0 to x_t



And, we note that

$$\lim_{t \uparrow \infty} \bar{\alpha}_t = \lim_{t \uparrow \infty} \sqrt{\prod_{i=1}^t \alpha_i} = 0$$

Direct Forward Links: *Explicit Expression*

$$x_0 \xrightarrow{\alpha_1} x_1 \xrightarrow{\alpha_2} \cdots \longrightarrow x_{t-1} \xrightarrow{\alpha_t} x_t \xrightarrow{\alpha_{t+1}} \cdots \xrightarrow{\alpha_T} x_T$$

What does happen in forward process?

Direct Forward Links

We can say that since

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \bar{\epsilon}_t$$

we have the direct forward conditional distributions as

$$Q(x_t | x_0) \equiv \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, 1 - \bar{\alpha}_t)$$

Learning Reverse Diffusion: *Reverse Processes*

$$x_0 \xleftarrow{P_{\mathbf{w},1}} x_1 \xleftarrow{P_{\mathbf{w},2}} \dots \xleftarrow{P_{\mathbf{w},t}} x_t \xleftarrow{P_{\mathbf{w},T}} x_T$$

In the reverse trajectory: *we start with $x_T \sim \mathcal{N}(0, 1)$ and go as*

$$x_{t-1} \sim P_{\mathbf{w}}(x_{t-1}|x_t, t)$$

What would be the marginal distribution in reverse trajectory at $T - 1$?

We can use marginalization to write

$$\begin{aligned} \hat{P}_{T-1}(x_{T-1}) &= \int P(x_{T-1}, x_T) dx_T \\ &= \int P_T(x_T) P_{\mathbf{w}}(x_{T-1}|x_T, T) dx_T \end{aligned}$$

Learning Reverse Diffusion: *Reverse Processes*

$$x_0 \xleftarrow{P_{\mathbf{w},1}} x_1 \xleftarrow{P_{\mathbf{w},2}} \dots \xleftarrow{P_{\mathbf{w},t}} x_t \xleftarrow{P_{\mathbf{w},T}} x_T$$

What if we go all the way back to 0?

$$\begin{aligned} \hat{P}_0(x_0) &= \int P(x_{0:T}) \prod_{t=1}^T dx_t \\ &= \int P_T(x_T) P_{\mathbf{w}}(x_{T-1}|x_T, T) \dots P_{\mathbf{w}}(x_0|x_1, 1) \prod_{t=1}^T dx_t \\ &= \int P_T(x_T) \prod_{t=1}^T P_{\mathbf{w}}(x_{t-1}|x_t, t) dx_t \end{aligned}$$

Learning Reverse Diffusion by *Maximum Likelihood*

$$x_0^j \xleftarrow{P_{\mathbf{w},1}} x_1 \xleftarrow{P_{\mathbf{w},2}} \dots \xleftarrow{P_{\mathbf{w},t}} x_t \xleftarrow{P_{\mathbf{w},T}} x_T$$

We want to see the same final distribution as our data

$$D_{\text{KL}} \left(\hat{P}_0 \| P_{\text{data}} \right) \approx 0$$

So, we need to **maximize** the **likelihood**

$$\begin{aligned} \log \mathcal{L}(\mathbf{w}) &= \sum_j \log \hat{P}_0(x_0^j) \\ &= \sum_j \log \int P_T(x_T) P_{\mathbf{w}}(x_0^j | x_1, 1) \prod_{t=2}^T P_{\mathbf{w}}(x_{t-1} | x_t, t) dx_t dx_1 \end{aligned}$$

Maximum Likelihood Learning

Maximum Likelihood on Reverse Trajectory

We learn reverse trajectory by maximizing the log-likelihood on our dataset

$$\max_{\mathbf{w}} \log \int P_T(x_T) \prod_{t=1}^T P_{\mathbf{w}}(x_{t-1}|x_t, t) dx_t$$

- + *Don't we care about the samples in between?!*
- Why should we?!

MLE Learning: *Training Objective*

Let us do some notations simplification: *we define*

$$P_{\mathbf{w}}(x_0) = \int P_T(x_T) \prod_{t=1}^T P_{\mathbf{w}}(x_{t-1}|x_t, t) \, dx_t$$

In MLE, we want to maximize

$$\log P_{\mathbf{w}}(x_0)$$

*This is however **computationally very hard!***

- + *Isn't this the same thing we had in VAE?!*
- *Right!*

MLE Training: *Finding an ELBO*

We first write the equation compactly as

$$P_{\mathbf{w}}(x_0) = \int \underbrace{P_T(x_T) \prod_{t=1}^T P_{\mathbf{w}}(x_{t-1}|x_t, t)}_{P_{\mathbf{w}}(x_{0:T})} dx_t = \int P_{\mathbf{w}}(x_{0:T}) dx_{1:T}$$

Now we do importance sampling: say we know good distribution $\Lambda(x_{1:T}|x_0)$

$$\begin{aligned} \log P_{\mathbf{w}}(x_0) &= \log \int P_{\mathbf{w}}(x_{0:T}) dx_{1:T} \\ &= \log \int \frac{P_{\mathbf{w}}(x_{0:T})}{\Lambda(x_{1:T}|x_0)} \Lambda(x_{1:T}|x_0) dx_{1:T} \\ &= \log \mathbb{E}_{x_{1:T} \sim \Lambda} \left\{ \frac{P_{\mathbf{w}}(x_{0:T})}{\Lambda(x_{1:T}|x_0)} \right\} \end{aligned}$$

Finding ELBO

Next we use Jensen's inequality to write that

$$\begin{aligned}\log P_{\mathbf{w}}(x_0) &= \log \mathbb{E}_{x_{1:T} \sim \Lambda} \left\{ \frac{P_{\mathbf{w}}(x_{0:T})}{\Lambda(x_{1:T}|x_0)} \right\} \\ &\geq \mathbb{E}_{x_{1:T} \sim \Lambda} \left\{ \log \frac{P_{\mathbf{w}}(x_{0:T})}{\Lambda(x_{1:T}|x_0)} \right\} = \text{ELBO}(\mathbf{w}|x_0)\end{aligned}$$

This describes an ELBO!

Implicit MLE via ELBO Maximization

We can maximize likelihood by maximizing the ELBO

- + *Shall we again think of Λ to be learned?!*
- *Not really! Actually we can explicitly compute a good Λ*

Posterior Calculation

The best $\Lambda(x_{1:T}|x_0)$ is given by *posterior*, i.e.,

$$\Lambda(x_{1:T}|x_0) = Q(x_{1:T}|x_0)$$

If we try to open up this posterior in reverse direction we have

$$\begin{aligned}\Lambda(x_{1:T}|x_0) &= Q(x_{1:T}|x_0) \\ &= Q(x_T|x_0) Q(x_{T-1}|x_T, x_0) \dots Q(x_{t-1}|x_{t:T}, x_0) \dots \\ &= Q(x_T|x_0) \prod_t Q(x_{t-1}|x_{t:T}, x_0)\end{aligned}$$

Posterior Calculation

We are interested in such distributions

$$Q(x_{t-1}|x_{t:T}, x_0) = Q(x_{t-1}|x_t, \mathbf{x}_{t+1:T}, x_0)$$

Let's do a bit of calculations

$$\boxed{Q(x_{t-1}|x_t, \mathbf{x}_{t+1:T}, x_0)} Q(\mathbf{x}_{t+1:T}|x_t, x_0) = Q(x_{t-1}, \mathbf{x}_{t+1:T}|x_t, x_0)$$

$$Q(x_{t-1}|x_t, \mathbf{x}_{t+1:T}, x_0) \prod_{i=t}^{T-1} Q(\mathbf{x}_{i+1}|x_i) = Q(x_{t-1}, \mathbf{x}_{t+1:T}|x_t, x_0)$$

Alternatively, we could say

$$Q(\mathbf{x}_{t+1:T}|x_{t-1}, x_t, x_0) Q(x_{t-1}|x_t, x_0) = Q(x_{t-1}, \mathbf{x}_{t+1:T}|x_t, x_0)$$

$$\prod_{i=t}^{T-1} Q(\mathbf{x}_{i+1}|x_i) \boxed{Q(x_{t-1}|x_t, x_0)} = Q(x_{t-1}, \mathbf{x}_{t+1:T}|x_t, x_0)$$

Posterior Calculation

So, the posterior can be simplified as

$$Q(x_{1:T}|x_0) = Q(x_T|x_0) \prod_t Q(x_{t-1}|x_t, x_0)$$

- + *How can we compute this?*
- *We can use the forward process*

We do know $Q(x_t|x_{t-1})$ and $Q(x_t|x_0)$: so we could write

$$\begin{aligned} Q(x_{t-1}|x_t, x_0) &= \frac{Q(x_{t-1}, x_t, x_0)}{Q(x_t, x_0)} \\ &= \frac{Q(x_{t-1}, x_t|x_0) P(x_0)}{Q(x_t|x_0) P(x_0)} = \frac{Q(x_{t-1}|x_0) Q(x_t|x_{t-1})}{Q(x_t|x_0)} \end{aligned}$$

Posterior Calculation: *Gaussian Posterior*

Recall that

$$Q(x_t|x_{t-1}) \equiv \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, 1 - \alpha_t)$$

$$Q(x_t|x_0) \equiv \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, 1 - \bar{\alpha}_t)$$

Thus, it is easy to show that

$$Q(x_{t-1}|x_t, x_0) \equiv \mathcal{N}(\eta_t(x_0, x_t), \rho_t^2)$$

for the mean and variance

$$\eta_t(x_0, x_t) = \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_{t-1}}(1 - \bar{\alpha}_t)}{1 - \bar{\alpha}_t}x_t$$

$$\rho_t^2 = \frac{(1 - \bar{\alpha}_{t-1})(1 - \alpha_t)}{1 - \bar{\alpha}_t}$$

ELBO Calculation

The ELBO is hence given by

$$\begin{aligned}
 \text{ELBO}(\mathbf{w}|x_0) &= \mathbb{E}_{\Lambda} \left\{ \log \frac{P_{\mathbf{w}}(x_{0:T})}{\Lambda(x_{1:T}|x_0)} \right\} \\
 &= \mathbb{E}_Q \left\{ \log \frac{P(x_T) \prod_{t=1}^T P_{\mathbf{w}}(x_{t-1}|x_t)}{Q(x_T|x_0) \prod_{t=2}^T Q(x_{t-1}|x_t, x_0)} \right\} \\
 &= \mathbb{E}_Q \left\{ \log \left[\frac{P(x_T)}{Q(x_T|x_0)} \prod_{t=2}^T \frac{P_{\mathbf{w}}(x_{t-1}|x_t)}{Q(x_{t-1}|x_t, x_0)} P_{\mathbf{w}}(x_0|x_1) \right] \right\} \\
 &= \mathbb{E}_Q \left\{ -\log \frac{Q(x_T|x_0)}{P(x_T)} - \sum_{t=2}^T \log \frac{Q(x_{t-1}|x_t, x_0)}{P_{\mathbf{w}}(x_{t-1}|x_t)} + \log P_{\mathbf{w}}(x_0|x_1) \right\} \\
 &= -D_{\text{KL}}(Q_{0 \rightarrow t} \| \mathcal{N}^0) - \sum_{t=2}^T D_{\text{KL}}(Q_{t-1 \leftarrow t, 0} \| P_{\mathbf{w}, t}) + \mathbb{E}_Q \{ \log P_{\mathbf{w}}(x_0|x_1) \}
 \end{aligned}$$

ELBO Maximization: *Sample Loss*

Our ultimate goal is to

$$\max_{\mathbf{w}} \text{ELBO}(\mathbf{w}|x_0)$$

which by dropping terms that do not depend on $\mathbf{w}$, is done by

$$\min_{\mathbf{w}} \sum_{t=2}^T D_{\text{KL}}(Q_{t-1 \leftarrow t, 0} \| P_{\mathbf{w}, t}) - \mathbb{E}_Q \{ \log P_{\mathbf{w}}(x_0|x_1) \}$$

Thus, we have the following sample loss

$$R(\mathbf{w}|x_0) = \sum_{t=2}^T D_{\text{KL}}(Q_{t-1 \leftarrow t, 0} \| P_{\mathbf{w}, t}) - \mathbb{E}_Q \{ \log P_{\mathbf{w}}(x_0|x_1) \}$$

ELBO Maximization: *Gaussian Reverse Process*

We typically consider a Gaussian reverse process

$$P_{\mathbf{w}}(x_{t-1}|x_t) \equiv \mathcal{N}(\mu_{\mathbf{w}}(x_t, t), \sigma_t^2)$$

The KL divergence in this case reads

$$D_{\text{KL}}(Q_{t-1 \leftarrow t, 0} \| P_{\mathbf{w}, t}) = \frac{1}{2} \left[\frac{\|\mu_{\mathbf{w}, t} - \eta_t\|^2}{\sigma_t^2} + d \left(\frac{\rho_t^2}{\sigma_t^2} + \log \frac{\rho_t^2}{\sigma_t^2} - 1 \right) \right]$$

and we have

$$\mathbb{E}_Q \{ \log P_{\mathbf{w}}(x_0|x_1) \} = -\frac{1}{2} \left[\frac{\|\mu_{\mathbf{w}, 1} - x_0\|^2}{\sigma_1^2} + d \log 2\pi\sigma_1^2 \right]$$

ELBO Maximization: *Gaussian Reverse Process*

We can hence define the final loss function to be

$$\begin{aligned} R(\mathbf{w}|x_0) &= \sum_{t=2}^T \frac{1}{\sigma_t^2} \|\mu_{\mathbf{w}}(x_t, t) - \eta_t\|^2 + \frac{1}{\sigma_1^2} \|\mu_{\mathbf{w}}(x_1, 1) - x_0\|^2 \\ &= \sum_{t=1}^T \frac{1}{\sigma_t^2} \|\mu_{\mathbf{w}}(x_t, t) - \eta_t\|^2 \end{aligned}$$

if we define

$$\eta_1 = x_0$$

This looks like a reconstruction!

*This motivated **DPM** and **DDPM** framework*